

TIANJIN BINHAI INTL, China

WMO: 545273

Lat: 39.124N

Lon: 117.346E

Elev: 10

StdP: 14.69

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.3	15.7	-16.9	2.2	28.4	-12.6	2.8	28.6	25.9	26.8	23.6	25.9	7.1	340	0.477

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.9	94.9	73.1	92.8	73.5	89.8	73.1	81.5	87.2	80.0	85.9	78.8	84.4	8.3	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.4	158.0	85.2	78.7	149.1	83.4	77.1	140.9	82.2	45.3	88.0	43.7	87.1	42.4	83.9	88.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.9	19.5	16.6	8.1	99.5	3.9	3.3	5.3	101.8	3.0	103.8	0.9	105.6	-1.9	108.0
			5.7	83.9	3.6	1.8	3.1	85.2	1.0	86.2	-1.1	87.2	-3.7	88.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	56.3	26.7	32.9	45.3	58.7	70.0	76.9	81.2	79.6	71.3	58.2	42.6	30.6	(6)
(7)		DBStd	20.08	5.77	6.42	7.70	7.05	6.05	5.07	4.25	4.18	5.46	6.48	7.41	5.35	(7)
(8)		HDD50	2252	721	480	188	11	0	0	0	0	0	11	239	602	(8)
(9)		HDD65	4905	1186	900	611	210	22	1	0	0	12	224	672	1067	(9)
(10)		CDD50	4549	0	0	42	273	619	806	969	917	639	266	18	0	(10)
(11)		CDD65	1726	0	0	1	22	176	356	504	452	201	14	0	0	(11)
(12)		CDH74	16493	0	0	11	253	1591	3338	5399	4376	1426	99	0	0	(12)
(13)		CDH80	6470	0	0	2	50	553	1380	2343	1707	421	14	0	0	(13)
(14)	Wind	WSAvg	7.2	6.9	7.2	8.6	9.5	8.9	7.6	6.6	5.8	6.1	6.5	6.6	6.7	(14)
(15)	Precipitation	PrecAvg	20.6	0.1	0.2	0.3	0.9	1.5	3.1	6.1	4.8	1.9	1.1	0.5	0.1	(15)
(16)		PrecMax	30.6	0.5	0.7	1.7	3.0	2.9	6.1	12.4	12.5	4.9	5.3	2.2	0.4	(16)
(17)		PrecMin	14.0	0.0	0.0	0.0	0.2	0.2	0.3	2.8	2.1	0.1	0.0	0.0	0.0	(17)
(18)		PrecStd	4.4	0.1	0.2	0.4	0.9	0.8	1.4	2.2	2.1	1.2	1.0	0.6	0.1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.8	57.2	73.8	85.8	94.7	97.1	98.9	95.2	91.2	81.0	67.6	50.1	(19)
(20)		MCWB	34.8	42.4	52.6	60.8	65.1	68.0	75.1	76.2	69.5	61.0	51.0	37.3	(20)	
(21)		2%	DB	42.8	52.1	68.2	80.6	89.5	93.5	95.2	92.8	87.7	76.8	62.6	46.1	(21)
(22)		MCWB	32.8	38.9	49.7	58.8	64.1	69.0	75.6	76.0	67.7	59.4	48.7	35.4	(22)	
(23)		5%	DB	39.4	48.3	63.0	76.8	86.0	91.0	93.0	89.9	84.5	73.0	58.6	42.8	(23)
(24)		MCWB	30.6	37.3	47.3	56.9	63.2	69.6	76.2	75.0	67.3	58.4	46.8	34.1	(24)	
(25)		10%	DB	37.0	44.5	58.9	73.0	82.4	87.7	90.0	87.9	82.2	69.6	54.0	39.6	(25)
(26)		MCWB	29.4	35.1	45.0	55.4	62.0	69.7	76.0	74.7	66.5	56.5	44.7	32.3	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.5	45.1	57.0	66.1	73.3	79.4	83.6	83.5	78.0	68.0	55.4	40.9	(27)
(28)		MCDWB	43.4	51.8	68.3	79.0	83.1	86.9	90.4	88.4	83.9	73.5	60.0	43.9	(28)	
(29)		2%	WB	33.9	41.2	52.6	63.0	70.2	76.5	82.0	81.5	74.9	65.2	51.5	37.6	(29)
(30)		MCDWB	40.0	47.6	63.5	74.0	80.7	84.0	87.9	86.6	79.7	70.6	57.7	42.0	(30)	
(31)		5%	WB	32.0	38.8	49.7	60.4	67.9	74.8	80.5	79.8	72.6	62.5	49.2	35.6	(31)
(32)		MCDWB	37.2	46.0	59.8	71.2	78.7	82.5	86.8	85.1	77.9	67.2	55.4	39.9	(32)	
(33)		10%	WB	30.2	36.3	46.7	58.0	65.8	73.2	79.4	78.5	70.5	59.9	46.7	33.8	(33)
(34)		MCDWB	35.5	43.4	55.7	67.9	76.2	80.8	85.3	83.6	76.7	66.2	52.2	38.3	(34)	
(35)	Mean Daily Temperature Range	MDBR	16.7	18.5	19.7	20.2	19.7	17.6	14.9	15.1	17.7	18.4	17.1	15.5	(35)	
(36)		5% DB	MCDBR	20.4	23.8	25.7	26.3	25.1	22.4	18.9	17.5	20.9	22.0	22.3	19.6	(36)
(37)		MCWBR	12.6	13.6	12.2	10.8	9.3	8.1	7.3	7.1	7.8	9.5	11.7	12.1	(37)	
(38)		5% WB	MCDBR	18.7	21.9	23.7	23.3	20.6	17.6	15.0	14.6	17.0	17.9	19.6	16.6	(38)
(39)		MCWBR	12.5	13.7	12.6	11.9	10.1	8.3	7.5	7.5	8.7	10.1	12.0	11.8	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.468	0.544	0.564	0.631	0.675	0.806	0.806	0.755	0.680	0.653	0.535	0.484	(40)	
(41)		taud	1.869	1.741	1.725	1.645	1.595	1.430	1.471	1.542	1.615	1.620	1.799	1.884	(41)	
(42)		Ebn at Noon	203	204	219	215	209	183	181	186	188	172	180	186	(42)	
(43)		Edh at Noon	46	61	68	78	83	99	94	85	75	68	50	43	(43)	
(44)	All-Sky Solar Radiation	RadAvg	751	975	1417	1686	1888	1686	1519	1477	1322	1011	764	665	(44)	
(45)		RadStd	76	114	127	89	128	162	111	91	108	100	107	73	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d) +1.01	(e) N/A	(f) N/A	(g) N/A	(h) -0.76	(i) -1.40	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(46)
(47)	Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+144	+110	(47)

Nomenclature: See separate page